

Workshop 3 .2 Turing Machine and decidable

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1. One way to show that the language $EDFA$ is decidable is to construct a Turing machine T that works as defined below.

$$EDFA = \{ \langle B \rangle \mid B \text{ is a DFA and } L(B) = \emptyset \}$$

Decider D On input $\langle B \rangle$: 1. Let $R := \{ q_0 \}$ (q_0 is the initial state) 2. Repeat: for every $q \in R$ and every $a \in \Sigma$, if there is a transition $\delta(q, a) = p$, add p to R 3. Stop when R no longer grows 4. If $R \cap F = \emptyset$ accept (language is empty); else, reject

a) Which states are in R when the loop ends in step 3?

- When the loop ends, R contains all reachable states from q_0

b) Explain why if $R \cap F = \emptyset$ then $L(B)$ is an empty set.

- Because there is no string that leads to the accepting state

c) Explain why D is a decider (i.e. why always halt).

- D always halts because it has finite states, so the set R only can grow finitely.

d) What is the time complexity of D when the input is an automaton with Q states, T transitions, and N symbols in the alphabet?

- because the loop is depending for states Q and number of characters N it will be $O(Q * N)$

2. Show that $ANFA = \{ \langle B, w \rangle \mid B \text{ is a NFA and } B \text{ accepts } w \}$ is decidable.

- If we convert the NFA to a DFA to simulate the string, we know that it's going to halt we read the entire string and if at the end its in an accepting state it accepts it otherwise dont.

3. Explain, in your own words, why we know that there exists languages that are not recognizable.

- We know that there are infinite languages but Turing machines are countable therefore we can say that there are more languages than Turing machines, and some languages are not recognized by any Turing machine, these languages are non recognizable languages.

4. Given the language:

$$ECFG = \{ \langle G \rangle \mid G \text{ is a CFG and } L(G) = \emptyset \}$$

And knowing that $ACFG = \{ \langle G, w \rangle \mid G \text{ is a CFG and } G \text{ derives } w \}$ is decidable

We propose the following argument to show that $ECFG$ PROOF IDEA:

Since $ACFG$ is decidable, there exists a TM A that decides $ACFG$. Let be $s_1, s_2, s_3, \dots$ the strings that belong to Σ^* , where Σ is the alphabet of G . Build a TM E that "on input $\langle G \rangle$: Enumerate $s_1, s_2, s_3, \dots$ one by one. For each s_i ($i \geq 1$) run A with input $\langle G, s_i \rangle$. If A accepts, reject. Else continue with the next string." WORKSHOP 3 Analyze the proof idea and answer:

a) Does this proof idea apply the concept of reduction? If so, which problem is reduced to which other problem? Justify your answer

- This idea doesn't use the concept of reduction, it only use A like a decider to solve E

b) Does this idea prove that $ECFG$ is decidable? Justify your answer

- It doesn't, because the grammar can generate an empty language, the machine never stops because it keeps checking for strings forever, so it's not a decider

5. Given the language

$$ETM = \{ \langle M \rangle \mid M \text{ is a TM and } L(M) = \emptyset \}$$

And the following idea to prove decidability of ETM :

We suppose ETM is decidable, therefore exists a TM E that decides ETM . Then, we can build a TM S that decides ATM (the acceptance problem) as follows: $S =$ "On input $\langle M, w \rangle$ 1) Use $\langle M \rangle$ and $\langle w \rangle$ to construct a TM M' . $M' =$ "On input x : // on any input If $x \neq w$, reject. If $x = w$, simulate M on input w and accept if M does." (Note: M' has M and w hardwired) 2) Run E on input $\langle M' \rangle$ 3) If E accepts, reject; if E rejects, accept. Analyze the proof idea and answer:

a) What strings could M' accept? Justify your answer

- M' only accepts the string w , this only happens if M accepts w .

b) What strings does M' reject? Justify your answer

- M' rejects every string different from w , this happens if M doesn't accept w .

c) What is the maximum size of $L(M')$?

- 1 because at most one string is accepted.

d) Explain why S answers to the question: Does M accepts w ?

- Because if M accepts w $L(M') \neq \emptyset$ then E rejects $\langle M' \rangle$ so S accepts.
If M doesn't accept w then $L(M') = \emptyset$ then E accepts $\langle M' \rangle$ so S rejects.
Therefore S correctly answered when M accepts w .

e) Does this proof idea apply the concept of reduction? If so, which problem is reduced to which other problem? Justify your answer

- Yes, it does

$$A_{TM} \leq E_{TM}$$

the idea tries to prove that E_{TM} is decidable therefore A_{TM} it is.

f) Provide the conclusion of the proof by concluding whether ETM is decidable or undecidable

- Because we know that A_{TM} is undecidable the prove for E_{TM} is decidable therefore A_{TM} it's impossible. So E_{TM} is undecidable.